

The Position Lagrangian: the Particle Energy and Its Certificates

Theorem monograph T05 of the chess-dynamics-lab program — the TORUS layer

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Keywords: Lagrangian, position energy, mobility, determinism

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Setting

The TORUS layer is the kinetic layer of the model: a particle move is a state transition, and the position carries energy. We define the Lagrangian energy of a position as the weighted sum of the potential (material) and kinetic (mobility) terms and prove its basic properties: computability, determinism, the initial certificate values and the upper bounds. This energy is what the full-game search (T10) minimizes and what the web laboratory displays as the energy panel.

The choice of weights answers two requirements: the energy must be integral in hundredths (for exact comparisons in the search) and must vanish on balanced symmetric positions. The material weight is 100 points per pawn unit, the kinetic coefficient is $\mu = 0.1$ moves per point: mobility matters but does not dominate the material — the classical proportion of chess practice, adopted as a definition.

Theorem

Theorem T05 (the position Lagrangian). Let $M(s)$ be the material of side s in pawn units, $m(s)$ the number of legal moves of side s . The energy is $E = [M(\circ) - M(\bullet)] + \mu[m(\circ) - m(\bullet)]$ with $\mu = 0.1$. Then: (i) E is exactly computable and deterministic; (ii) in the initial position $m(\circ) = m(\bullet) = 20$; (iii) after 1.e4 the White mobility equals 30; (iv) for any position with the standard material set of one side (queen, two rooks, two bishops, two knights, king) the mobility of that side does not exceed $27 + 14 + 14 + 13 + 13 + 8 + 8 + 8 = 105$.

Proof

(i) Computability and determinism. The material $M(s)$ is a finite sum over the 64 squares of a fixed function of the square content; the mobility $m(s)$ is the cardinality of the finite set of legal moves produced by a deterministic generation procedure. Both quantities are independent of the traversal order and of any random parameters, so E is a deterministic function of the position. The correctness of the move generation is certified by the perft identities of T08.

(ii) The initial mobility. The eight pawns of the second rank have two moves each (one and two squares forward) — 16 moves; both knights (b1 and g1) have two moves each (a3, c3 and f3, h3) — 4 more; the bishops, rooks, queen and king are blocked by the own pawns. Altogether exactly $16+4 = 20$ legal moves, which matches $\text{perft}(1)$ of Theorem T08. By symmetry $m(\bullet) = 20$.

(iii) After 1.e4. The e2 pawn advanced to e4 and still has two moves; the seven other pawns give 14; the f1 bishop diagonal opened (five squares: e2, d3, c4, b5, a6), the d1 queen diagonal opened (four squares: e2, f3, g4, h5) and the square e2 became available to the g1 knight (a third move beyond f3 and h3). In total: pawns $7 \cdot 2 + 2 = 16$, knights $2 + 3 = 5$, the f1 bishop 5, the queen 4; the sum is $16 + 5 + 5 + 4 = 30$. Note that the count follows the blocking rule and every listed square is indeed free after 1.e4.

(iv) The upper bound. The mobility of a side does not exceed the sum of the maximal mobilities of its particles: every move is made by exactly one particle, and the number of moves of a particle does not exceed its empty-board mobility (blocking can only reduce the number of reachable squares). By T03 the maxima are: queen 27, rook 14, bishop 13, knight 8, king 8. For the standard set (queen, two rooks, two bishops, two knights, king) this gives $27 + 2 \cdot 14 + 2 \cdot 13 + 2 \cdot 8 + 8 = 105$. We note that with promoted material the actual record of mobility in a legal position reaches 218 moves — this constant enters the protocol data as the~“maximal kinetic energy” reference of the eight-queen construction. The theorem is proved.

Protocol data

Energy certificates (check C7)

Quantity	Value
White mobility, initial position	20
White mobility after 1.e4	30
Energy determinism	confirmed (double run)
Zobrist hash stability	confirmed
splitmix64 invertibility	confirmed (2000 values)
Mobility upper bound (standard set)	105
Maximal mobility reference (218 construction)	218

Summary of what is proved

We introduced and justified the Lagrangian energy of a position: computability and determinism are proved, the initial certificates 20 and 30 are computed with the full enumeration of the contributing moves, the upper bound 105 for the standard material set is established, and the 218 reference of the queen construction is fixed. The mechanism is direct combinatorial counting with the blocking rule; all values enter protocol C7 and the baseline.

Verification

Check: `python3 dynamics.py -run C7` or the C7 line of the polyglot core. The determinism is verified by a double energy evaluation and a comparison of the Zobrist hashes; the mobilities 20 and 30 are compared with the move generation; PASS in all seven implementations.